

DEEP LEARNING PROJECT

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DATASETS

- **train.csv** – Historical sales data for 1,115 stores from 2013 to 2015 to train our models.
- **store.csv** – metadata containing sales information.
- **test.csv** – to test model performance
- **Reason:** These datasets provide sales history, store features, and information for future predictions.

CLEANING THE DATA

- Converted date column to datetime format
- Created cyclical encodings for DayOfWeek, Month and DayInYear to capture seasonal patterns.
- Filtered out closed stores and records with Sales = 0
- Cleaned StateHoliday to ensure consistent type
- Imputed missing values in CompetitionDistance, CompetitionOpenSinceYear/Month.
- Imputed Promo2 related missing values and standardized PromoInterval.



FEATURE ENGINEERING

Lag & Rolling Features:

- Created Sales and Customers lag features for 7, 14, and 28 days.
- Rolling mean features over 7 and 28 days.

Competition & Promo2 Features:

- CompetitionOpenMonths calculated from store open date and current date.
- Promo2Active and Promo2Season flags based on promotional schedule.

Encoding:

- Categorical Columns were encoded (Store, StoreType, Assortment, PromoInterval, StateHoliday.)

Scaling:

- StandardScaler applied to numeric features to normalize data.

DATA DROPPED

- Dropped Open column after filtering closed stores
- Dropped original Date column ,it was replaced with engineered features
- Dropped duplicated or irrelevant data

TYPES OF
NETWORKS
USED

- DNN (Dense Neural Network): Simple feedforward network, predicts sales .
- LSTM (Long Short-Term Memory): Captures temporal dependencies (sales trends, seasonality).
- GRU (Gated Recurrent Unit): Lightweight alternative to LSTM for time dependencies

ARCHITECTURE OVERVIEW

Model	Architecture & Dropout	Notes
DNN	$200 \rightarrow 100 \rightarrow 50 \rightarrow 1$	Fully connected, linear output
LSTM (baseline)	$\text{LSTM}_{128} \rightarrow \text{Dense}_{64} \rightarrow \text{Dense}_{32} \rightarrow 1$, Dropout 0.2	Best baseline model
LSTM (tuned)	$\text{LSTM}_{64} \rightarrow \text{Dense}_{64} \rightarrow \text{Dense}_{32} \rightarrow 1$, Dropout 0.1	LR=0.005, batch=64
GRU	$\text{GRU}_{64} \rightarrow \text{Dense}_{64} \rightarrow \text{BatchNorm} \rightarrow \text{Dense}_{32} \rightarrow 1$, Dropout 0.3	Slightly worse than LSTM

Overview

Data

Code

Models

Discussion

Leaderboard

Rules

Team

Submissions

Submission and Description

Private Score ⓘ

Public Score ⓘ

Selected

 **sub_dnn.csv**
Complete (after deadline) · 2h ago

1.00000

1.00000

☐

 **submission_final_dnn.csv**
Error (after deadline) · 2h ago

 **submission_final_dnn.csv**
Error (after deadline) · 2h ago

 **submission_final_dnn.csv**
Error (after deadline) · 2h ago

 **submission.csv**
Complete (after deadline) · 16h ago

0.59596

0.60451

☐

THANK YOU

Asha Deen